



Multi-AZ was chosen over a Read Replica for LogiHaul's RDS MySQL instance because the primary requirement is automatic failover, not read scaling. LogiHaul's relational workload driver profiles, shipper accounts, and invoice records is write-sensitive and financially critical. If the primary RDS instance in AZ-1 fails, Multi-AZ promotes the standby in AZ-2 automatically, typically within 60–120 seconds, with no manual intervention and no data loss, because the standby is kept synchronously in sync with the primary at all times.

A Read Replica was rejected for this role because it uses asynchronous replication meaning there is always a small replication lag, and promotion to primary in a failure event requires manual action. For a logistics billing system where an unconfirmed invoice means an unpaid driver, that combination of lag and manual promotion is not acceptable.

The 50x traffic spike LogiHaul is designed for hits the order-intake path (DynamoDB, Lambda), not the billing layer so the higher read throughput a Read Replica provides is not needed here. Multi-AZ gives exactly what this tier requires: durability and automatic recovery, at a cost that stays within LogiHaul's \$150/month steady-state budget.